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Paleomagnetism of Late Mesozoic Pelagic Limestones From the Southern Alps

J. VANDENBERG

Department of Structural and Applied Geology and the Paleomagnetic Laboratory, State University of Utrecht, Utrecht, Netherlands

A. A. H. WONDERS

Department of Stratigraphy and Micropaleontology, State University of Utrecht, Utrecht, Netherlands

Paleomagnetic data are presented from the Southern Alps (Belluno-Bassano del Grappa). These paleomagnetic data specify the megatectonic movements of the Southern Alps for the Late Mesozoic. The detailed paleomagnetic data of the upper Albian, Cenomanian, and Turonian show upper Albian reversals and the possible presence of a short mixed interval in the Cenomanian.

INTRODUCTION

Channell and Tarling [1975] started the research in the Southern Alps on pelagic sediments, and they first recognized differences between paleomagnetic data from the Southern Alps and from the Central Apennines. Further paleomagnetic research [*VandenBerg and Wonders,* 1976] was carried out and showed that the movement pattern of the Southern Alps was closely related to the movements of Africa and of peninsular Italy. Discrepancies in amounts of rotation between the Southern Alps and the Central Apennines were explained by differential movements of both crustal blocks in the Late Cretaceous and Tertiary. Although these paleomagnetic data were rather convincing, we felt that they had to be backed up by still more intensive studies.

The aim of the present study was to specify in detail the changes in declination and inclination of paleomagnetic directions in continuous sections of Cretaceous pelagic carbonate rocks. Therefore we chose a very close sampling of restricted intervals. Since our earlier work had shown that megatectonic movements were to be expected in the Mid-Cretaceous, we mainly directed our attention to the Upper Albian, Cenomanian, and Turonian limestones. In this way it became possible besides to obtain information about polarity changes and to compare them to the results from the Gubbio section [*Lowrie and Alvarez,* 1977; *Alvarez et al.,* 1977].

SAMPLING LOCALITIES

Two sections were studied. One is located close to Bassano del Grappa and is referred to as the 'Quero-Schievenin' section (see Figure 1). The other section is not far from Belluno and is called the 'Valle del Mis' section (see Figure 1).

Quero-Schievenin Section

This section ranges from Triassic to Maastrichtian and is composed of gently folded rocks, while at some places, normal faults are present. Only the Upper Cretaceous part was studied. The Cenomanian is exposed as white, well-stratified pelagic limestones with black chert lenses and the Turonian to Maastrichtian as pink-to-red, pelagic limestones with rare chert. The whole section contains rich planktonic foraminiferal faunas. The studied interval (Cenomanian to Early Campanian) is some 100 m thick, which means that the sedimentation rate was about 5 m/m.y. No hiatus seems to be present.

The *globotruncanoides* and *cushmani* zones of the Cenomanian were sampled at 36 levels, several per level. The position of the levels was entirely determined by the absence of chert intercalations. The upper part (Turonian to Early Campanian) was sampled with a larger spacing: one site every 5 m (14 sites and 82 samples). This part of the section was studied before [*VandenBerg and Wonders,* 1976].

Valle del Mis Section

Upper Triassic to Maastrichtian rocks are exposed in subvertical position in the valley of the river Mis. Only the upper part of this section was sampled. The Albian is developed as a sequence of thin-bedded siliceous limestones with nodules, lenses, and beds of chert. The limestones are fossiliferous micrites and biomicrites, with a sedimentation rate of about 7m/m.y. The sedimentation rate drops abruptly toward the end of the Albian. The Cenomanian is represented by about 50 cm of pink marly limestone. The Turonian (about 10 m thick) consists of pink-to-red marly limestones with occasional siliceous beds. Higher stratigraphic units are badly exposed, with exception of the Maastrichtian, which is composed of marly limestones. From the Cenomanian to the Maastrichtian this section is strongly reduced in comparison with other sections [*Casati and Tomai,* 1969].

The upper Albian (about 19 m thick) was sampled at 55 levels. The reduced Turonian sequence was sampled at 100 levels. The whole sequence contains rich planktonic foraminiferal faunas and could be dated appropriately.

MAGNETIC PROPERTIES AND LABORATORY TREATMENT

Every core was sliced into cylinders (2.5-cm diameter and 2.2-cm height) at Fort Hoofddijk Laboratory. All measurements were carried out on a two-axis ScT Cryogenic magnetometer. The natural remanent magnetization (NRM) intensities for the white limestone specimens from the Valle del Mis section ranged from 0.2 to 2.0 $\times$ 10^{-3} A/m, while the white specimen from the Quero-Schievenin section revealed somewhat lower intensities: 0.02–1.0 $\times$ 10^{-3} A/m.

The red-to-pink limestones appeared to have appreciable higher NRM intensities: in the Valle del Mis section, 7.0–30.0 $\times$ 10^{-3} A/m, and in the Quero-Schievenin section, 1.0–20.0 $\times$ 10^{-3} A/m.

From both sections about 20 pilot specimens were demagnetized progressively in alternating fields in about 10 steps, the white limestone specimens in the range from 0 mT to 50 mT and the reddish limestone specimens from 0 mT to

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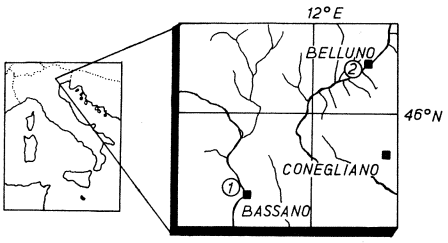


Fig. 1. Map showing the geographic position of the Quero-Schievenin section (1) and the Valle del Mis section (2) in northern Italy (Southern Alps).

300 mT alternating field peak values. The magnetic behavior was studied by means of orthogonal projection figures of the vector path during demagnetization. Secondary magnetization components in the white limestone specimens were removed in alternating fields above 5 mT, and for the reddish specimens above 20 mT. This is in good accordance with the behavior observed on pelagic carbonates from Umbria [*VandenBerg et al.*, 1978]. Therefore all white limestone specimens were demagnetized in four to six steps in alternating fields with peak values of 5 mT to 25 mT. The reddish specimens were treated in fields of 20 mT to 80 mT. In Figure 2, representative examples are given of the magnetic behavior of white and reddish limestone specimens upon demagnetization.

Results and Their Interpretation

In Figures 3*a*, 3*b*, 4*a*, and 4*b* the results on sample level are compiled. The declination and inclination of the characteristic primary components, after cleaning and after correction for the bedding tilts, are plotted versus the stratigraphic position in these figures. On the left-hand side the planktonic foraminiferal zones are given, as they could be recognized in thin sections.

The paleomagnetic directions from the two sections gave for the corresponding intervals a positive fold test, and therefore the results of both sections were combined on basis of the assigned ages. In Table 1 the mean results are given applying Fisherian statistics.

Interpretation of the Magnetostratigraphy

Valle del Mis section. The lower part of this section of late Albian age (Figure 3*b*) reveals constant values of declination and inclination throughout the 19-m sequence. At the base and near the top, reversed magnetic directions are present, indicating at least two reversals in the upper Albian. In paleomagnetic studies on cores of Deep Sea Drilling Project site 263 [*Green and Brecher,* 1974; *Jarrard,* 1974], reversals in the same interval have been reported but could not be dated appropriately. Only 1 m above the upper reversal the declination reaches the value of the previous interval, showing a rather slow recovery of the geomagnetic field direction.

The upper part of the section, late Cenomanian and Turonian in age, shows an appreciable larger scatter in declination and inclination (Figure 3*a*). No reversals seem to be present.

Quero-Schievenin section. The lower part of this section (Figure 4*b*) represents biostratigraphically most of the Cenomanian and turned out to be of particular interest. The lower 12 m revealed rather constant declination and inclination values, but at the top an interval (estimated to represent about

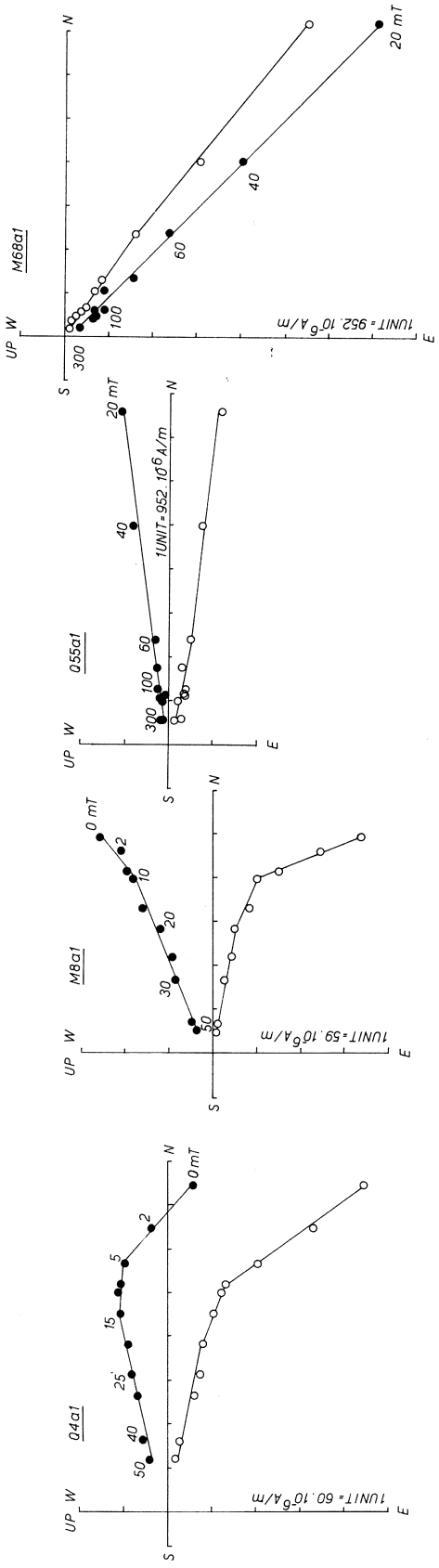


Fig. 2. Orthogonal projection diagrams of the vector path of the total NRM vector during af demagnetization. Open circles denote projections on the vertical N-S plane; solid circles are projections on the horizontal plane. Numbers represent values of the applied alternating peak fields in milliteslas. Samples Q4a1 and M8a1 are from white pelagic limestones, whereas Q55a1 and M68a1 are from reddish limestones.

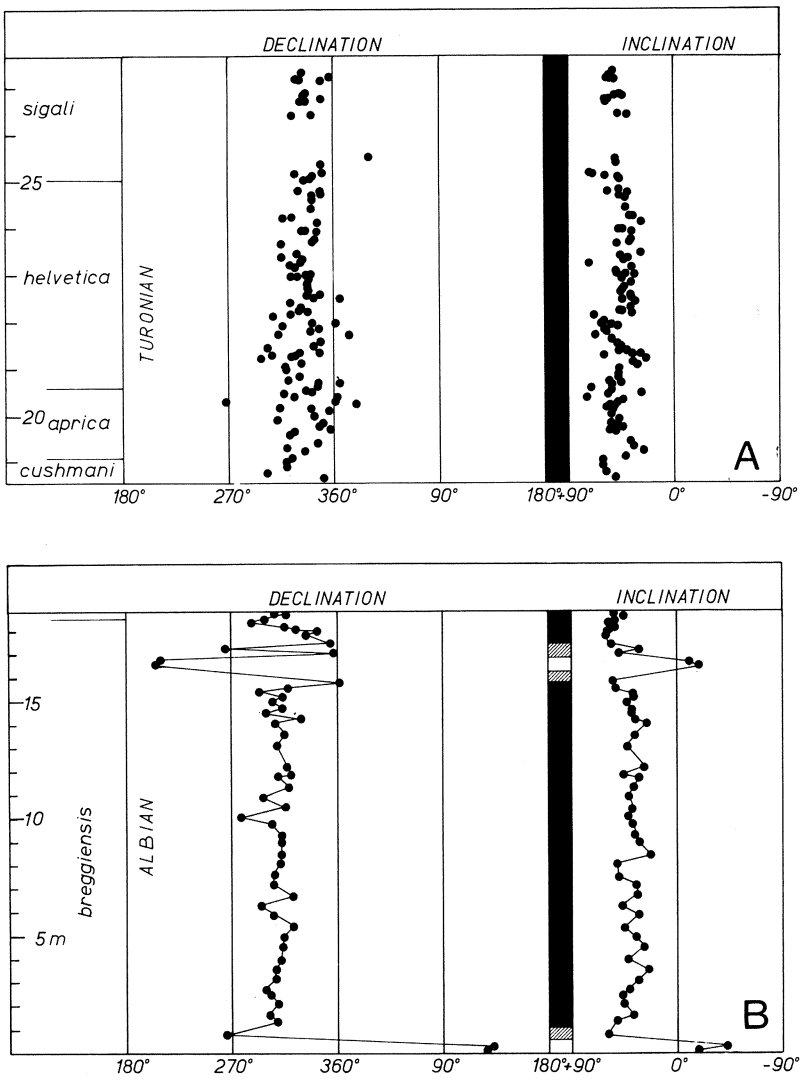


Fig. 3. Sequence of sample directions plotted versus stratigraphic position (*a*) for the upper part of the Valle del Mis section and (*b*) for the lower part. Planktonic foraminiferal zones are given in the left column.

1.3 m.y.) revealed erratic directions. In this interval, reversed (at three levels), normal, and intermediate magnetic directions are present. No lithologic change was observed in that interval, neither a different magnetic behavior of the samples taken from that interval upon demagnetization. However, until the present no reversals or a period of important instability of the earth's magnetic field has been reported from other Cenomanian sections [*Lowrie and Alvarez,* 1977; *Channell et al.,* 1978], but that can very well be explained by the closer spacing of the samples from this study. Research is in progress to establish the presence of this mixed interval in other Cenomanian sections.

The upper part of this section (Figure 4*a*) covers a longer time span: Turonian to early Campanian. The results of our earlier study are incorporated in Figure 4*a* as open squares. The declination and inclination values are very constant until the upper three sites. It can be observed that below and above the only site with reversely magnetized samples, intermediate paleomagnetic directions are present. The age assignment makes it highly probable that these magnetic directions are related with the Gubbio 'A' reversal [*Lowrie and Alvarez,* 1977; *Alvarez et al.,* 1977].

Interpretation of Movements With Respect to the Pole

When the directional variations of a paleomagnetic record of a sequence are to be interpreted in terms of megatectonic movements with respect to the pole, all changes of the geomagnetic field itself have to be balanced out. Changes in the geomagnetic field due to secular variation are assumed to be of minor importance, because it is averaged out during relatively long time of acquisition of the NRM, related to the low sedimentation rate of pelagic carbonates. Therefore only long-term variations of the paleomagnetic field (more over 10,000 years) can possibly be traced in these sediments. The durations of polarity transitional zones were apparently long enough for pelagic sediments with sedimentation rates of 5m/m.y. or more. The mean values given in Table 1 were calculated from only those samples that were close to the overall mean value of the interval concerned, and the intermediate

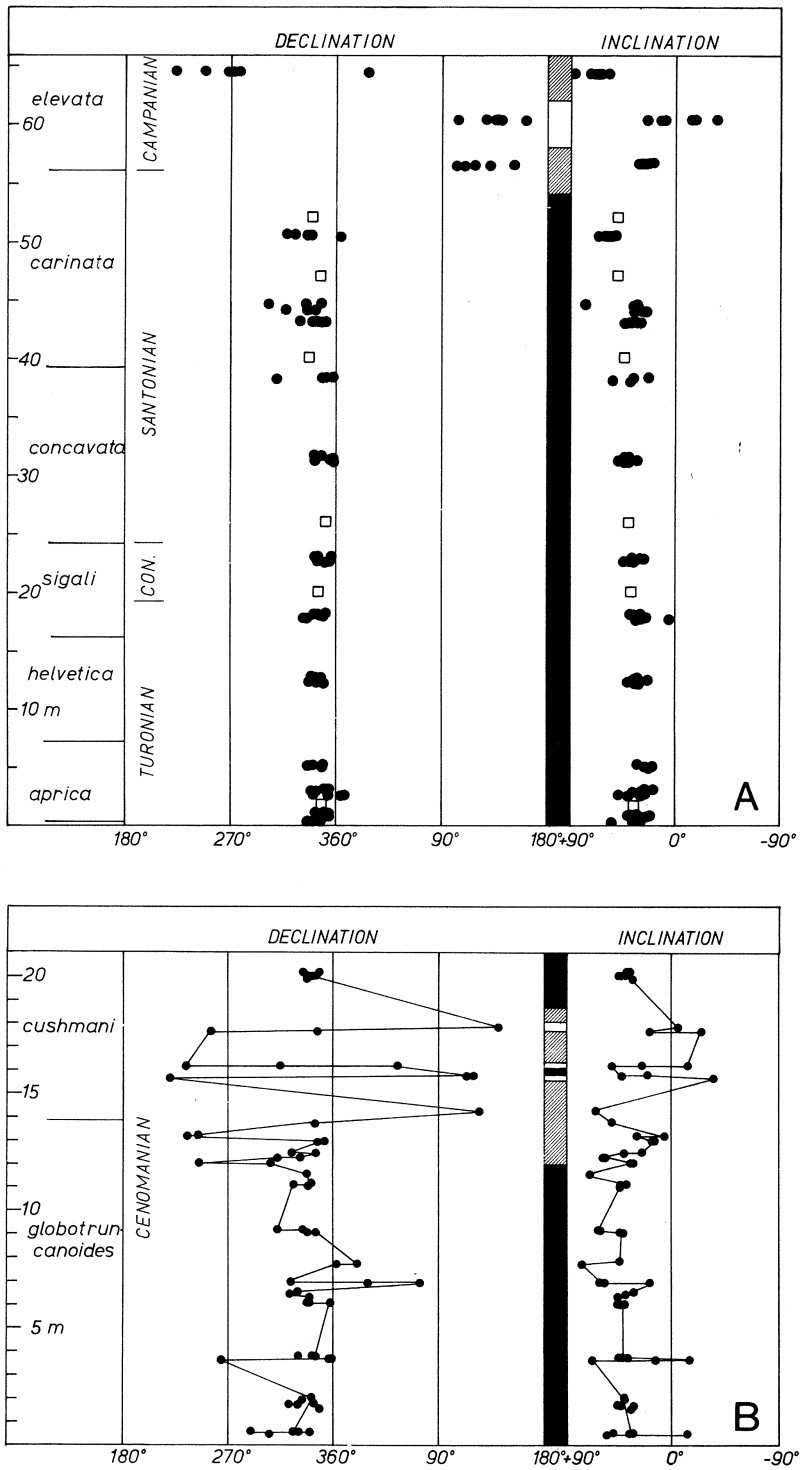


Fig. 4. Sequence of sample directions plotted versus stratigraphic position (*a*) for the upper part of the Quero-Schievenin section and (*b*) for the lower part. Planktonic foraminiferal zones are in the left column of the figure. Open squares represent earlier results [*VandenBerg and Wonders,* 1976].

results from the *elevata* and *cushmani* zones of the Quero-Schievenin section were therefore omitted. These results allow an interpretation about megatectonic movements of the Southern Alpine block and entirely support earlier conclusions [*VandenBerg and Wonders,* 1976].

In Figure 4*b* can be observed the slight change to higher inclination values, but this is more pronounced and substantiated in the mean results of Table 1. Apparently, the Southern Alpine crustal block remained at the same latitude during late Albian to Turonian times and moved northward

TABLE 1. Paleomagnetic Results

Planktonic Foraminiferal Zone	Age	Mean Direction f, deg: Declination	Mean Direction f, deg: Inclination	a_{95}	N	n	Pole Position	dp	dm
Elevata	early Campanian	347.6	+50.0	...	2	11	72.0°N, 228.7°E	...	...
Carinata	Santonian	346.7	+49.4	8.3	3	17	71.1°N, 230.0°E	7.3	11.0
Concavata	late Coniacian	343.4	+45.7	8.4	3	16	67.0°N, 232.6°E	6.8	10.7
Sigali	L. Turonian/early Coniacian	345.1	+42.9	7.4	8	50	65.7°N, 226.7°E	5.6	9.1
Helvetica	Turonian	343.8	+39.5	5.5	11	66	62.9°N, 226.6°E	3.9	6.5
Aprica	early Turonian	340.4	+37.7	9.5	8	50	60.4°N, 231.4°E	6.6	11.2
Globotruncanoides	early Cenomanian	336.4	+36.2	4.6	8	44	57.6°N, 236.7°E	3.1	5.3
Breggiensis	late Albian	325.8	+38.2	5.4	8	48	52.9°N, 252.3°E	3.7	6.3

N is number of 'sites'; n is number of samples; f, giving unit weight to each site.

later in the Cretaceous. The inclination values of the Albian to Turonian data are lower than those of Late Triassic and Jurassic age, implying a southward movement during Early Cretaceous. These very same movements could be interpreted from the paleomagnetic data of Umbria [*VandenBerg et al.*, 1978].

Declination values of about 330° in upper Paleozoic and Early Mesozoic rocks remained constant until upper Albian times. Data from Late Cretaceous rocks show an increase of about 15°. The mean declination value of the Late Cretaceous and Early Tertiary rocks is about 345° (Figure 5 and Table 1). This implies an anticlockwise rotation of about 15° during latest Albian and Cenomanian. A further 15° counterclockwise rotation of post-Early-Tertiary age and an additional northward movement are necessary to reach the present declination and inclination value of the geomagnetic field.

The deduced two-phase movement scheme is in good accordance with that for the Central Apennines [*VandenBerg et al.*, 1978], as far as the timing of the movements is concerned. The rotation values, however, do differ appreciably as was noted before [*Channell and Tarling*, 1975; *VandenBerg and Wonders*, 1976]. The conclusion that differential movements must have occurred between the Southern Alpine block and the Central Apennine block during the Late Cretaceous and Tertiary is confirmed by the present data.

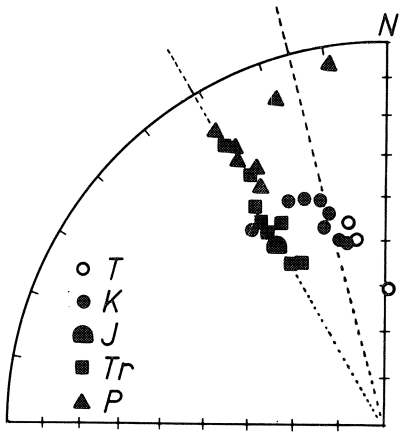


Fig. 5. Mean paleomagnetic directions for the Southern Alps in equal-area plot (lower hemisphere) of the presently available data. The Permo-Triassic and Tertiary data are from *Zijderveld and Van der Voo* [1973]; the Jurassic data are from *VandenBerg and Wonders* [1976]; the Cretaceous data are from the present paper (Table 1).

CONCLUSIONS

The deviation toward the west of paleomagnetic declinations from the Southern Alps was caused by two subsequent rotations of about 15°, the first during late Albian to Cenomanian and the second during the Tertiary.

Upper Albian reversals and the possible presence of a mixed interval in the Cenomanian show that in the Cretaceous quiet interval from Aptian to the early Campanian not only a normal polarity magnetic field has existed.

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REFERENCES

Alvarez, W., M. A. Arthur, A. G. Fischer, W. Lowrie, G. Napoleone, I. Premoli-Silva, W. M. Roggenthen, Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy, V, Type section for the Late Cretaceous–Paleocene geomagnetic reversal time scale, *Geol. Soc. Amer. Bull., 88,* 383–389, 1977.

Casati, P., and M. Tomai, Il Giurassico ed il Cretacico del versante settentrionale del Vallone Bellunese e del gruppo del M. Brandol, *Riv. Ital. Paleontol., 75,* 205–340, 1969.

Channell, J. E. T., and D. H. Tarling, Palaeomagnetism and the rotation of Italy, *Earth Planet. Sci. Lett., 25,* 177–188, 1975.

Channell, J. E. T., W. Lowrie, F. Medizza, and W. Alvarez, Palaeomagnetism and tectonics in Umbria, Italy, *Earth Planet. Sci. Lett., 39,* 199–210, 1978.

Green, K. A., and A. Brecher, Preliminary paleomagnetic results for sediments from site 263, leg 27, in *Initial Reports of the Deep Sea Drilling Project,* vol. 27, pp. 405–413, U. S. Government Printing Office, Washington, D. C., 1974.

Jarrard, R. D., Paleomagnetism of some leg 27 sediment cores, in *Initial Reports of the Deep Sea Drilling Project,* vol. 27, pp. 415–423, U. S. Government Printing Office, Washington, D. C., 1974.

Lowrie, W., and W. Alvarez, Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy, III, Upper Cretaceous magnetic stratigraphy, *Geol. Soc. Amer. Bull., 88,* 374–377, 1977.

VandenBerg, J., and A. A. H. Wonders, Paleomagnetic evidence of large fault displacement around the Po-Basin, *Tectonophysics, 33,* 301–320, 1976.

VandenBerg, J., C. T. Klootwijk, and A. A. H. Wonders, The Late Mesozoic movements of the Italian peninsula: Further paleomagnetic data from the Umbrian sequence, *Geol. Soc. Amer. Bull., 89,* 133–150, 1978.

Zijderveld, J. D. A., and R. Van der Voo, Paleomagnetism in the Mediterranean area, in *Implications of Continental Drift to Earth Sciences,* vol. 1, edited by D. H. Tarling and S. K. Runcorn, pp. 133–161, Academic, New York, 1973.

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